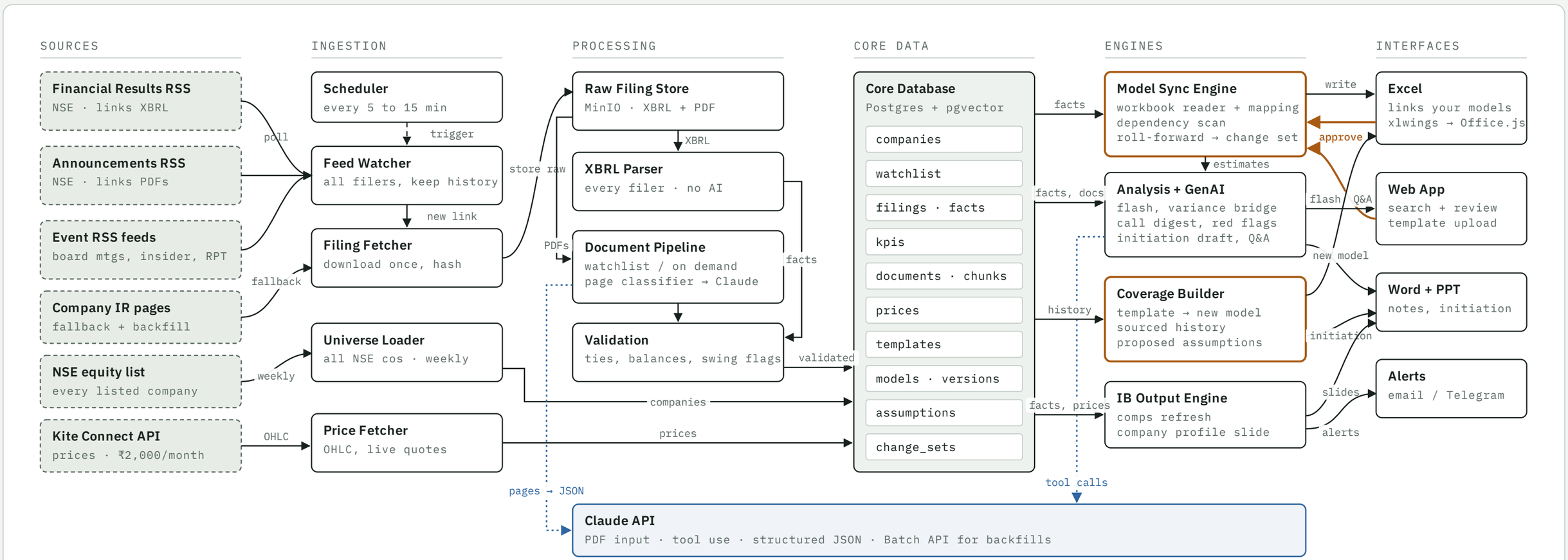
SYSTEM DESIGN V1.0.1 · OCTOBER 2026

Analekh Architecture

The analyst's record, kept for you.

Every NSE-listed company is preloaded and its results are parsed as they are filed. For a company you follow, the system finds and updates your Excel model after you approve each change, or builds a new model from your sector template and drafts an initiation. Core numbers come from XBRL; the AI writes the analysis and pulls every figure from the database.

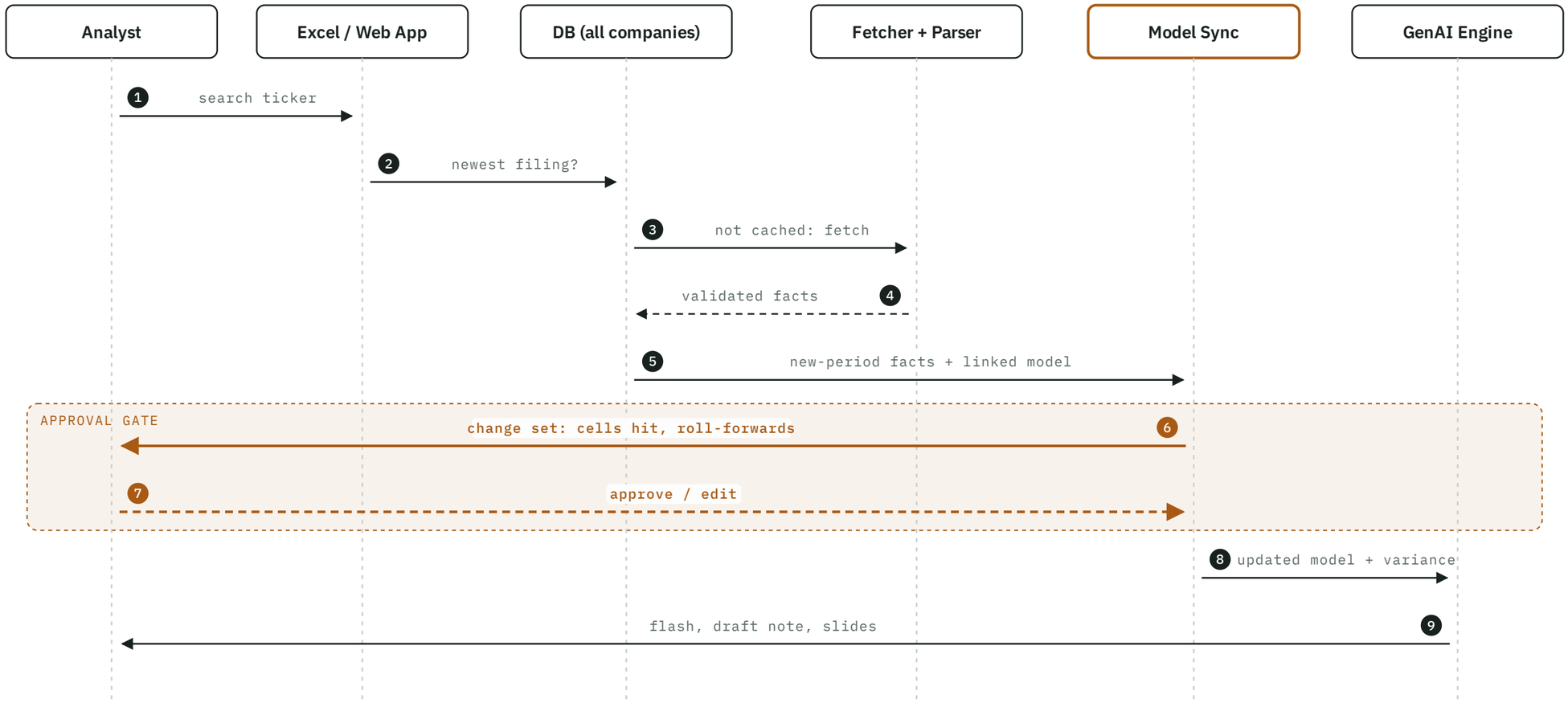
Components and data flow



data · control / return · analyst approval · Claude API call · external source

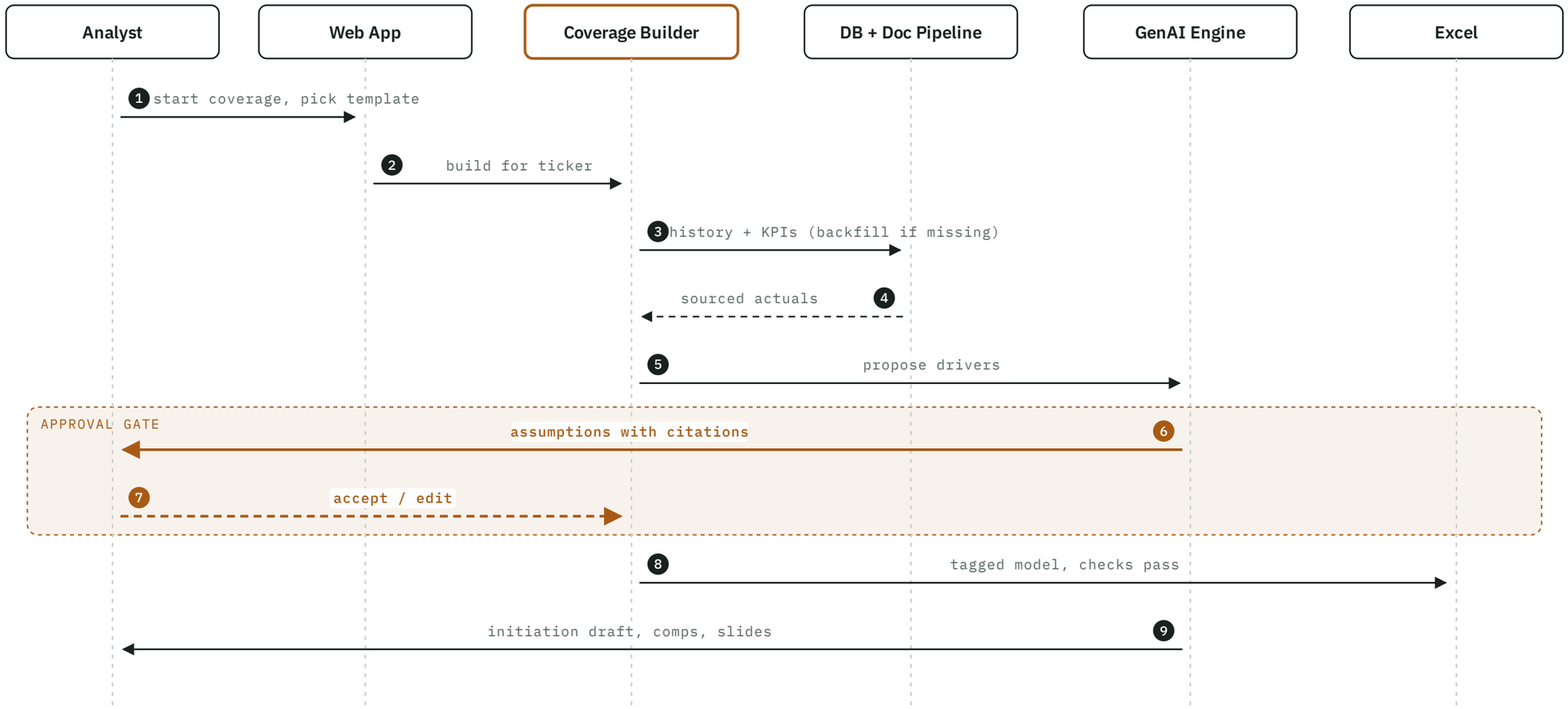
The NSE equity list preloads every company and NSE's RSS feeds are polled on a schedule, so results for all filers are parsed automatically and "newest filing for this ticker" is a database lookup. PDFs go through a local page classifier and only the relevant pages reach Claude, for watchlist companies or on demand. Model Sync updates existing models; the Coverage Builder creates new ones from uploaded sector templates. Nothing is written to Excel until the analyst approves.

What happens when an analyst enters a ticker



Steps 3 and 4 run only when the filing isn't already in the database. The linked model is found through watched folders or the open workbook, identified by a hidden tag, its file name or its contents. Steps 6 and 7 are the approval gate. The analyst sees which cells change and which forecasts need rolling forward, accepts or edits each one, and only then is a new model version saved.

What happens when an analyst starts coverage



Used when no model exists yet. The analyst picks a company and a sector template (their own upload or a built-in one). The builder fills the template's inputs with sourced history and keeps its formulas. Proposed assumptions pass through the same approval gate before the tagged model is saved. From the next results onward, the quarterly update flow above takes over.

Numbers come from data

Figures in drafts are fetched from the database or the model through tool calls, and each one carries its XBRL tag or PDF page reference.

Data rights

Use NSE's published RSS feeds at their stated refresh interval and company IR pages. Don't call NSE's internal web endpoints. Get written confirmation from NSE or license a feed before any public release.

Your templates

Upload a model template per sector. The system reads its structure, you confirm the mapping once, and a test build must pass before it is used.

Runs from India

NSE's main site is geo-restricted, so the scheduler and fetchers run on an Indian network or a VM in an Indian region.

Stack

`Python 3.12 · FastAPI · Celery + Redis · Arelle · PyMuPDF · Claude API · Postgres + pgvector · MinIO · xlwings · React · Docker Compose`